

PHD

ARITHMETIC EXPRESSIONS

Chapter Summary + 40 Practice Questions

Current NCERT focus: understanding expressions through values, terms, brackets, equivalent forms and reasoning - not merely applying a memorised order-of-operations rule.

SUMMARY OF THE MAIN POINTS

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| 1. Arithmetic expressions | An arithmetic expression is a mathematical phrase formed using numbers and operations such as $+$, $-$, $\times$ and $\div$. Every expression has a value. |
| 2. Equality and equivalent expressions | Different expressions can have the same value. The equality sign ($=$) can show that two expressions are equivalent. |
| 3. Comparing expressions | Expressions can often be compared using $<$, $>$ or $=$ by reasoning about how their parts differ, without fully calculating both values. |
| 4. Brackets control meaning | Brackets show which part of an expression must be treated together. For example, $5 \times (8 + 2)$ is different from $5 \times 8 + 2$. |
| 5. Terms | Terms are the parts of an expression separated by addition signs. Subtraction can be rewritten as addition of the additive inverse; for example, $18 - 7 = 18 + (-7)$. |
| 6. Swapping and grouping terms | When an expression is written as a sum of terms, the terms may be swapped or regrouped without changing the value. These are the commutative and associative properties of addition. |
| 7. Evaluating complex expressions | First evaluate expressions inside brackets. Products and quotients that form a term are evaluated before the terms are added together. |
| 8. Removing brackets | If a bracket is preceded by $+$, signs inside stay the same. If it is preceded by $-$, the signs of the terms inside change when the bracket is removed. |
| 9. Distributive property | A multiple of a sum or difference can be distributed: $a \times (b + c) = a \times b + a \times c$ and $a \times (b - c) = a \times b - a \times c$. |
| 10. Efficient calculation | Equivalent expressions can make arithmetic easier, for example $98 \times 25 = (100 - 2) \times 25$. |

Useful habit: Before calculating, ask: What are the terms? What is inside brackets? Can I rewrite the expression in an easier but equivalent form?



40 PRACTICE QUESTIONS

Work neatly. Where a question asks for reasoning, try to avoid unnecessary long calculations.

A. Foundations and Equivalent Expressions

1. Write two different arithmetic expressions whose value is 18.
2. Fill in the blank: $27 + 8 = \underline{\hspace{2cm}} + 10$.
3. Fill in the blank: $54 - \underline{\hspace{2cm}} = 39$.
4. Fill in the blank: $7 \times \underline{\hspace{2cm}} = 84 \div 2$.
5. Six notebooks cost ₹28 each. Write an arithmetic expression for the total cost and find its value.
6. A customer pays ₹500 for items costing ₹135 and ₹86. Write one expression using brackets for the change received, then find the change.
7. Explain the difference between the expression $18 + 7 \times 3$ and the equation $18 + 7 \times 3 = 39$.
8. Arrange these expressions in ascending order of value: $72 - 19$, $50 + 6$, 7×8 , $162 \div 3$.

B. Compare by Reasoning

9. Insert $<$, $>$ or $=$ without fully evaluating both sides: $435 + 198 \underline{\hspace{2cm}} 436 + 196$.
10. Insert $<$, $>$ or $=$: $723 - 158 \underline{\hspace{2cm}} 724 - 159$.
11. Insert $<$, $>$ or $=$: $307 + 489 \underline{\hspace{2cm}} 309 + 486$.
12. Insert $<$, $>$ or $=$: $620 - 247 \underline{\hspace{2cm}} 618 - 244$.
13. Insert $<$, $>$ or $=$ and explain briefly: $25 \times 19 \underline{\hspace{2cm}} 25 \times 20$.
14. Compare by reasoning: $36 \times (18 - 7) \underline{\hspace{2cm}} 36 \times (17 - 6)$.

C. Terms and Evaluating Expressions

15. Identify the terms in: $38 - 7 + 4 \times 6$.
16. Identify the terms in: $45 - 3 \times 8 + 20 \div 5$.
17. Rewrite using addition of additive inverses: $62 - 19 + 7 - 12$.
18. Find the value by thinking in terms: $28 - 7 + 8$.
19. Find the value: $39 - 2 \times 6 + 11$.
20. Find the value: $48 - 10 \times 2 + 16 \div 2$.

D. Brackets and Sign Changes

21. Find the value of $30 + 5 \times 4$ by first identifying its terms.
22. Find the value of $(30 + 5) \times 4$. Explain in one sentence why it differs from Question 21.
23. Remove the brackets without changing the value: $75 + (18 - 9)$.
24. Remove the brackets without changing the value: $120 - (35 + 12)$.
25. Remove the brackets without changing the value: $200 - (70 - 25)$.
26. Add brackets to make the statement true: $48 - 12 + 6 = 30$.
27. Add brackets to make the statement true: $60 - 20 - 10 = 50$.

E. Distributive Property and Smart Calculation

28. Expand using the distributive property: $7 \times (20 + 3)$.
29. Write as a single product by taking out the common factor: $9 \times 17 + 9 \times 3$.
30. Evaluate efficiently using a nearby round number: 98×24 .
31. Evaluate efficiently: 103×17 .
32. Evaluate efficiently: 49×36 .
33. Given that $53 \times 18 = 954$, find 63×18 without starting the multiplication from the beginning.

F. Expressions in Real Situations

34. Five friends buy one sandwich each at ₹42 and together leave a ₹20 tip. Write an expression for the total amount, identify its terms and find the value.
35. A museum ticket costs ₹50 for an adult and ₹30 for a child. Write and evaluate an expression for the cost of tickets for 3 adults and 4 children.
36. A hall has 8 rows of 14 chairs and 6 extra chairs near the stage. Write an expression and find the total number of chairs.
37. A shopkeeper receives ₹500 for two items costing ₹145 and ₹238. Write an expression using brackets and find the change.
38. A shop packs 96 kg of rice into 3 kg bags and already has 5 such bags ready. Write an expression for the total number of bags and find its value.

G. Reasoning Challenges

39. Without fully calculating each expression, identify all expressions equal to $84 - (29 + 11)$:
(A) $84 - 29 - 11$ (B) $84 - 29 + 11$ (C) $84 + (-29) + (-11)$ (D) $84 - (29 - 11)$.
40. Start with the number 12. One student writes $(12 + 8) \times 3$; another writes $12 \times 3 + 8$. Find both values, explain why they are different, and write a real-life situation that could match each expression.

Reference alignment: NCERT Ganita Prakash, Class 7 Mathematics, Part I, Chapter 2 - Arithmetic Expressions. Practice questions in this worksheet are newly written and are not copied from the textbook.